Stony Brook Institute at Anhui University

安徽大学纽约石溪学院

Chapter 5 Large Number Law and Central Limit Theorem

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Law of Large Numbers (LLN)	Chebyshev's LLN/weak LLN
	Bernoullis's LLN
	Khinchin's LLN

Central Limit Theorem (CLT)	Lindberger-Levy CLT
	De Moirve-Laplace CLT
	Liapunov CLT

Introduction

- The Law of Large Numbers shows broadly that the frequency and some average of many independent observations are **stable and predictable**.

- While the Central Limit Theorem says that the distribution of a sum or average of many small random quantities can be **approximated** by **Gaussian distribution .**

Background of LLN

Stability of average results in a large number of random phenomena:

The frequency of heads in a large number of coin flips

Reject rate during production

Letter frequency

Chebyshev's law of large numbers

Chebyshevs inequality

The Chebyshevs inequality provides a bound estimation on the probability of a random variable departing from its mean by a certain amount.

Theorem(Chebyshevs Inequality) If X is a random variable with finite mean μ and variance σ^2 , then for any value $\varepsilon > 0$,

$$P\{|X-\mu| \geq \varepsilon\} \leq \frac{\sigma^2}{\varepsilon^2}$$

or

$$\mathrm{P}\{|X-\mu| < \varepsilon\} \geq 1-\frac{\sigma^2}{\varepsilon^2}$$

Chebyshev's law of large numbers

Example

Suppose we flip a fair coin 1000 times. How likely it is to obtain between 40% and 60% heads?

Solution Let the r.v. $Xi(i = 1; 2; \cdots; 1000)$ be 1 if the coin lands on head and 0 otherwise, i.e.

$$X_i = \begin{cases} 1 & \text{if } i^{th} \text{ outcome is head} \\ 0 & \text{if } i^{th} \text{ outcome is tail} \end{cases}$$

then, $\mu = E(Xi) = 0.5$,

$\sigma^2 = Var(Xi) = 1/4$;

Chebyshev's law of large numbers

By Chebyshevs inequality, the probability to obtain between 40% and 60% heads can be estimated by :

$$
\begin{aligned}
&\mathrm{P}\left\{1000 \times 40\% < X_1 + X_2 + \cdots + X_{1000} < 1000 \times 60\%\right\} \\
&= \mathrm{P}\left\{400 < X_1 + X_2 + \cdots + X_{1000} < 600\right\} \\
&= \mathrm{P}\left\{\frac{4}{10} < \frac{X_1 + X_2 + \cdots + X_{1000}}{1000} < \frac{6}{10}\right\} \\
&= \mathrm{P}\left\{\left|\frac{X_1 + X_2 + \cdots + X_{1000}}{1000} - \frac{1}{2}\right| < \frac{1}{10}\right\} \\
&\geq 1 - \frac{1/4}{1000(1/10)^2} = 0.975
\end{aligned}
$$

Chebyshev's law of large numbers

Notes

As Chebyshevs inequality is valid for all r.v. X with finite mean and variance, we cannot expect the estimated bound on the probability will be very close to the actual probability in most cases. That means Chebyshevs inequality usually gives **very loose bounds** on probability.

Example

Consider a normal random variable with mean μ and variance σ^2 , the following table shows the probability bound by Chebyshevs inequality and the true value.

Probability	Chebyshev's bound	$X \sim N(\mu, \sigma^2)$
$P(\lvert X-\mu\rvert \geq \sigma)$	≤ 1	0.3173
$P(\lvert X-\mu\rvert \geq 2\sigma)$	$\leq 1/2^2 = 0.25$	0.0465
$P(\lvert X-\mu\rvert \geq 3\sigma)$	$\leq 1/3^2 = 0.11$	0.00270
$P(\lvert X-\mu\rvert \geq 4\sigma)$	$\leq 1/4^2 \approx 0.06$	0.000063

Chebyshev's law of large numbers

Law of Large Numbers (LLN)

Let X_1, X_2... are sequence of independent random variables, they all have finite variance, and the variance has a common upper bound, that is, D(Xi) ≤ K, i =1, 2... , for any ε>0:

$$\lim_{n\to\infty} P\left\{\left|\frac{1}{n}\sum_{i=1}^{n} X_i - \frac{1}{n}\sum_{i=1}^{n} E(X_i)\right| < \varepsilon\right\} = 1$$

Arithmetic mean

Mathematical expectation

- The deviation of the arithmetic mean and the mathematical expectation is small, with a probability close to 1.
- When *n* is sufficiently large, The probability arithmetic mean gets close to its mathematical expectation is close to 1.

Chebyshev's law of large numbers

Special case of Chebyshev's law of large numbers

Theorem(Weak Law of Large Numbers) If random variables $X_1, X_2, \cdots, X_n, \cdots$ are independent and with mean and finite variance

$$E(X_n) = \mu, Var(X_n) = \sigma^2, n = 1, 2, \cdots$$

then

$$\lim_{n\to\infty} \mathrm{P}\left(\left|\frac{1}{n}\sum_{k=1}^{n} X_k - \mu\right| < \varepsilon\right) = 1 \qquad \forall \varepsilon > 0$$

This LLN is known as **the weak law of large numbers**, **or the law of averages**. It says broadly that the average results of many independent random variables are stable and predictable.

Chebyshev's law of large numbers

$$\lim_{n\to\infty} P\left\{\left|\overline{X_n}-\mu\right|<\varepsilon\right\}=\lim_{n\to\infty} P\left\{\left|\frac{1}{n}\sum_{k=1}^{n} X_k-\mu\right|<\varepsilon\right\}=1$$

Proof:

$$E\left(\frac{1}{n}\sum_{k=1}^{n} X_k\right)=\frac{1}{n}\sum_{k=1}^{n} E(X_k)=\frac{1}{n}\cdot n\mu=\mu$$

$$D\left(\frac{1}{n}\sum_{k=1}^{n} X_k\right)=\frac{1}{n^2}\sum_{k=1}^{n} D(X_k)=\frac{1}{n^2}\cdot n\sigma^2=\frac{1}{n}\sigma^2$$

Chebyshev's inequality:

$$(P(X-E(X)\geq\varepsilon)\leq\frac{D(X)}{\varepsilon^2}$$

Chebyshev's law of large numbers

Chebyshev's inequality:

$$(P(X - E(X) \geq \varepsilon) \leq \frac{D(X)}{\varepsilon^2}$$

$$P\left\{\left|\frac{1}{n}\sum_{k=1}^{n} X_k - \mu\right| < \varepsilon\right\} \geq 1 - \frac{\frac{\sigma^2}{n}}{\varepsilon^2}$$

Let $n \to \infty$, and then notice the probability ≤ 1, so:

$$\lim_{n\to\infty} P\left\{\left|\frac{1}{n}\sum_{k=1}^{n} X_k - \mu\right| < \varepsilon\right\} = 1$$

Chebyshev's law of large numbers

Weak law of large numbers

It can be used for a petrol station to decide how many tons of gasoline to stock and a fast-food restaurant deciding how many steamed buns to prepare even though many of their customers make independent decisions. This is because that the LLN says that these many individual decisions will produce a stable result.

Under the same assumption of the theorem above, we have

$$\mathrm{P}\left(\lim_{n\to\infty}\frac{1}{n}\sum_{k=1}^{n}X_k=\mu\right)=1$$

In words, it states that the arithmetic mean converges to the mathematical expectation with probability 1 as the sample size tends to infinity. As a special case of the above LLN, one can find the following theorem.

Bernoulli's LLN

Theorem(Bernoullis LLN) Let n_A be the number of the event A in n Bernoulli trials, and p be the probability of the event A on any one Bernoulli trial, then

$$\lim_{n\to\infty} \mathrm{P}\left(\left|\frac{n_A}{n} - p\right| < \varepsilon\right) = 1 \qquad \forall \varepsilon > 0$$

The Bernoulli's LLN shows the steadiness of the relative frequency of the events in large number of random trials.

Bernoulli's LLN

Theorem(Bernoullis LLN)

$$\lim_{n\to\infty} \mathrm{P}\left(\left|\frac{n_A}{n} - p\right| < \varepsilon\right) = 1 \qquad \forall \varepsilon > 0$$

- This means that the relative frequency n_A/n of the event A converges to the probability p of A with probability 1, i.e. n_A/n almost always converges to p.

- Therefore, in application, while the trial's times n is large enough, we can use the frequency to replace the probability of an event. This will help us to understand the precise meaning of frequency definition of probability.

- **Example:** If you toss a fair coin many, many times, the fraction of heads will approach 1/2.

Bernoulli's LLN

Question

Weak LLN and **Bernoullis LLN** are independent and identically distributed laws of large numbers, and both require the existence of the expectation and variance of random variables.

If there is no condition for the existence of variance of random variables, is the conclusion of LLN still valid?

Khinchin's LLN

Theorem(Khinchin's Law of Large Numbers) Let $X_1, X_2, \cdots, X_n, \cdots$ be a sequence of independent and identically distributed random variables having finite mean $\mu = E(X_i), i = 1, 2, 3 \ldots$. Let $\overline{X} = \frac{1}{n}\sum_{i=1}^{n} X_i$, then

$$P(lim_{n\to\infty}\overline{X} = \mu) = 1$$

Khinchin's LLN provides a practical way to find the expected value of r.v.

Summary

- LLN describes the arithmetic mean and frequency stability of samples from various angles.
- Provides a theoretical basis for the methods commonly adopted by people:

- Replace or estimate the mean value with the arithmetic mean of the sample;
- To substitute or estimate its "probability" with frequency.